

IWAKUNI MCAS, Japan

WMO: 477640

Lat: 34.145N

Lon: 132.247E

Elev: 6

StdP: 101.25

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 43324

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.9	0.2	-8.2	1.9	3.4	-7.0	2.1	3.6	10.0	5.9	9.1	6.4	2.8	300	0.400

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.0	32.9	26.4	31.8	25.9	30.8	25.7	27.2	31.0	26.6	30.3	26.0	29.8	4.1	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	21.5	29.8	25.5	20.7	29.4	24.9	20.0	29.0	86.2	31.2	83.4	30.3	80.8	29.9	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	7.8	6.8	DB	-2.6	34.7	1.5	1.0	-3.7	35.5	-4.5	36.1	-5.4	36.7	-6.4	37.4	(4)
(5)			WB	-4.2	28.4	1.3	0.9	-5.2	29.0	-5.9	29.6	-6.6	30.1	-7.6	30.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.4	5.8	6.2	9.4	14.3	19.0	22.5	26.7	27.9	24.3	19.0	13.3	8.0	(6)
(7)		DBStd	8.05	2.30	2.49	2.93	2.99	2.16	2.09	2.30	1.74	2.38	2.75	3.08	2.76	(7)
(8)		HDD10.0	372	130	110	48	4	0	0	0	0	0	0	6	74	(8)
(9)		HDD18.3	1644	387	341	278	123	17	1	0	0	0	24	153	321	(9)
(10)		CDD10.0	2717	1	2	28	133	280	376	518	556	428	279	105	12	(10)
(11)		CDD18.3	948	0	0	0	3	39	126	259	297	178	44	2	0	(11)
(12)		CDH23.3	7841	0	0	0	7	98	506	2491	3325	1281	132	1	0	(12)
(13)		CDH26.7	2368	0	0	0	0	6	62	806	1225	262	7	0	0	(13)
(14)	Wind	WSAvg	2.9	3.1	3.1	3.1	2.8	2.6	2.3	2.6	2.8	3.0	3.2	3.0	3.2	(14)
(15)	Precipitation	PrecAvg	1692	64	74	134	186	185	293	225	127	180	109	81	42	(15)
(16)		PrecMax	2603	169	205	292	354	375	521	696	472	392	288	204	104	(16)
(17)		PrecMin	1131	5	11	13	56	54	58	5	3	4	25	2	2	(17)
(18)		PrecStd	317	41	43	68	64	81	114	142	90	90	66	49	28	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.0	15.6	19.6	24.1	27.2	30.2	34.0	34.3	31.8	27.9	22.1	17.1	(19)
(20)			MCWB	9.4	11.5	13.2	16.5	19.3	24.2	26.3	26.7	25.4	22.8	17.4	13.0	(20)
(21)		2%	DB	12.1	13.2	17.8	21.9	25.7	28.2	32.8	33.2	30.2	26.1	21.0	15.2	(21)
(22)			MCWB	8.2	9.4	12.9	15.6	18.9	23.2	26.2	26.6	24.8	21.1	16.6	11.1	(22)
(23)		5%	DB	11.1	11.9	16.0	20.3	24.2	27.0	31.4	32.0	29.1	24.9	19.7	14.0	(23)
(24)			MCWB	7.2	8.2	11.9	15.2	18.4	22.4	25.9	26.0	24.1	19.8	15.8	10.4	(24)
(25)		10%	DB	10.0	10.8	14.2	19.1	23.2	26.0	30.3	31.1	28.1	23.8	18.4	12.9	(25)
(26)			MCWB	6.6	7.2	10.6	14.7	18.0	21.8	25.6	25.8	23.4	19.0	14.7	9.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	12.6	15.5	18.7	21.7	25.0	28.0	28.2	26.7	23.7	19.6	14.3	(27)
(28)			MCD B	12.8	14.1	17.5	21.8	24.5	28.5	31.7	32.3	30.1	26.5	21.2	16.1	(28)
(29)		2%	WB	9.3	10.3	13.8	17.4	20.6	24.1	27.1	27.5	25.7	22.5	18.0	12.2	(29)
(30)			MCD B	11.2	12.2	16.2	19.9	23.6	27.1	30.7	31.4	29.1	25.0	19.9	14.2	(30)
(31)		5%	WB	7.9	9.0	12.5	16.5	19.7	23.5	26.5	26.8	25.0	20.9	16.6	10.7	(31)
(32)			MCD B	10.2	11.0	15.4	19.1	23.0	26.1	30.2	30.6	28.0	23.7	18.8	13.3	(32)
(33)		10%	WB	6.7	7.7	11.0	15.5	18.8	22.6	25.9	26.3	24.3	19.6	15.2	9.7	(33)
(34)			MCD B	9.5	10.0	14.0	18.4	22.1	25.1	29.7	30.1	27.3	22.8	17.9	12.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.2	7.3	7.9	7.9	7.6	6.0	5.7	6.0	6.5	7.4	7.7	7.3	(35)
(36)			MCD BR	8.3	8.5	9.6	9.6	9.3	7.4	7.1	7.2	6.9	7.9	8.1	8.3	(36)
(37)			MCWBR	5.9	6.4	6.8	6.1	5.5	4.1	3.6	3.7	3.7	4.7	5.5	5.9	(37)
(38)		5% WB	MCD BR	7.7	7.4	8.5	7.6	7.5	6.2	6.5	6.0	6.4	7.1	7.4	(38)	
(39)			MCWBR	6.1	6.3	6.8	5.6	5.2	3.8	3.7	3.5	4.6	5.7	5.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.406	0.448	0.510	0.539	0.540	0.557	0.557	0.527	0.487	0.441	0.418	0.400	(40)	
(41)		taud	2.141	2.051	1.903	1.886	1.949	1.989	2.037	2.126	2.196	2.228	2.200	2.173	(41)	
(42)		Ebn at Noon	773	780	764	764	769	753	750	766	779	782	756	752	(42)	
(43)		Edh at Noon	121	148	185	197	187	180	170	153	137	122	114	112	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.19	2.88	3.92	4.81	5.29	4.58	4.92	5.10	4.09	3.40	2.49	1.94	(44)	
(45)		RadStd	0.18	0.24	0.30	0.44	0.54	0.49	0.67	0.51	0.41	0.34	0.20	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.37	N/A	+1.02	N/A	N/A	N/A	N/A	N/A	+113	N/A
(47) Regional (32 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page